

Build a Permanent Magnet Motor

A complete, interactive engineering plan derived directly from Howard R. Johnson's 1979 patent — covering theory, materials, fabrication, assembly, tuning, and operation. Every step, every part, every dimension you need.

Patent Facts

Number	US 4,151,431
Inventor	H. R. Johnson
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Claims	25
Drawings	10

Classification

International (Int. Cl.)

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U.S. Classification

310/12

310/152

Examiner
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What This Motor Does

The core inventive concept in one paragraph

The Johnson permanent magnet motor produces relative motion between an armature and a stator using only static magnetic fields — no electrical current, no commutator, no brushes. The unpaired electron spins within permanent magnets are treated as a continuous source of motive power, analogous to a room-temperature superconductor.

The breakthrough is geometric. The armature magnet's length is set to be slightly greater than the combined width of two stator magnets plus one inter-magnet gap. At every position along the track, the leading pole of the armature is repelled by an adjacent like pole while the trailing pole is attracted by an opposite pole. The resultant force vector always points in the same direction — producing continuous motion along the track.

Both a linear embodiment and a rotary embodiment are described. The rotary version includes a threaded shaft that allows the builder to axially displace the armature and thereby regulate the rotational speed — a fully mechanical throttle with zero electrical components.

Prototype Dimensions

From the patent's working example

Armature magnet length Slightly greater than $2 \times$ stator width + gap	3.000 in
Stator magnet width Perpendicular to track direction	1.000 in
Stator magnet thickness Pole-to-pole dimension	0.250 in
Stator magnet length Along track direction	4.000 in
Stator magnet grade Alnico V or modern NdFeB equivalent	Grain-oriented
Air gap (armature $\leftrightarrow$ stator) Tune for max force vs. min pulsation	≈ 0.125 in
Spacing between stator magnets Slightly varied per pair for smoothing	≈ 0.250 in

The armature length must be slightly longer than 2 stator widths + 1 gap. This is the entire trick.
Reversing the armature magnet (N $\leftrightarrow$ S) reverses the direction of motion. No electrical switching required.
Mechanical advantage claimed: greater than 100:1.

At a Glance

- 2 embodiments (linear + rotary)
- ~14 build steps total
- 19 BOM items
- 10 workshop tools
- 25 patent claims



Theory of Operation

How static magnetic fields produce continuous thrust — the physics behind the patent



Spin → Field → Force

Unpaired Electron Spins in a Ferromagnet → Source of the Permanent Magnetic Field

Unpaired Electron Spins in a Ferromagnet → Source of the Permanent Magnetic Field



1

Unpaired Electron Spins as a Power Source

Ferromagnetic materials (iron, nickel, cobalt, neodymium compounds) contain unpaired electrons in their atomic structure. Unlike paired electrons whose magnetic moments cancel, these unpaired electrons spin around the nucleus and produce a measurable magnetic field.

In Johnson's view, a permanent magnet behaves like a room-temperature superconductor: the spinning electrons never cease, and the flux they generate is available to do work without any external electron flow through a conductor.

Conventional motors switch the field via commutators, brushes, or AC current. In this patent, switching is achieved purely geometrically — by concentrating the stator flux and shaping the armature magnet so attraction and repulsion forces combine into a unidirectional thrust.

2 The Critical Geometry

The armature magnet length is slightly greater than the combined width of TWO stator magnets plus ONE inter-magnet gap. This is the geometric heart of the design — it guarantees that at every position the armature sits over a magnetic environment that is asymmetric in the direction of intended motion.

At every position, the leading pole of the armature experiences a strong repulsion from the same-polarity stator pole just ahead, while the trailing pole experiences a strong attraction from the opposite-polarity stator pole just behind. The resultant vector always points in the same direction along the track.

When the armature is reversed ($N \leftrightarrow S$), the direction of motion reverses. No electrical switching is required — the geometry alone does the work.

3 Flux Concentration via High-Permeability Backing

A plate of Netic Co-Netic (or equivalent μ -metal) is bonded directly to the south poles of the stator magnets. This high-permeability alloy offers a low-reluctance return path for flux, effectively concentrating the field on the north-pole side facing the armature.

Without this backing, a substantial fraction of the stator flux would leak out the back of the magnets and never reach the air gap. With the backing, flux density in the gap can rise by 30-60% (per equivalent μ -metal measurements), increasing the force on the armature proportionally.

The same principle is applied to the armature by bowing it (concave side down toward track) and beveling the pole ends. Both measures concentrate flux at the pole faces — exactly where the stator field is strongest.

4 Smoothing via Staggering & Variable Spacing

If only one armature magnet is used, the force vector pulses as the magnet crosses each stator gap. The patent mitigates this by using multiple armature magnets (2-3) staggered along the direction of motion so the impulses overlap and smooth out.

A second smoothing technique is to vary the spacing between adjacent stator magnets slightly. Because each armature pole sees a slightly different phase of the force cycle, the resultant thrust becomes more uniform along the track.

There is a trade-off: too small an air gap increases pulsation; too large a gap reduces force. The patent specifies an air gap of approximately 1/8 in (0.125 in) as a starting point; fine-tuning is required per build.

5 Speed Regulation by Axial Displacement

In the rotary embodiment, the armature is mounted on a threaded shaft. Rotating the knob advances or retracts the armature along the rotation axis, changing the overlap between the armature magnets and the stator magnets.

Greater axial overlap → stronger tangential force → higher RPM. Less overlap → weaker force → lower RPM. This provides a purely mechanical throttle with no electrical components.

Because the force scales non-linearly with overlap area, the calibration curve is steep near full engagement; small knob rotations produce significant speed changes near the top end. The builder should mark a reference scale on the knob.



Bill of Materials

Every part you need — magnets, structural, hardware, and tooling

Acquisition progress

0 / 19

All Categories

Magnet

Metal

Structural

Hardware

Tooling



Search parts, specs, sources...

✓	ID	PART	SPECIFICATION	QTY	PURPOSE	SOURCE
☐	BOM-01	Stator Magnets — Rectangular Blocks	Grade N42-N52 NdFeB, grain-oriented, 1 in × 0.25 in × 4 in (W × T × L)	8 (linear) / 12 (rotary)	Define the stator track of common polarity facing the armature	K&J Magnetics / CMS Magnetics / SuperMagnetMan
☐	BOM-02	Armature Magnets — Elongated Bar	Grade N42-N52 NdFeB, 3 in long, bowed (arcuate) profile, beveled pole ends	1 (single) or 2-3 (staggered)	Moving magnet that experiences attraction + repulsion to produce motive force	K&J Magnetics (custom cut) / Master Magnetics
☐	BOM-03	Rotary Armature Magnets — Wedge	Same NdFeB grade, wedge profile with curved inner surface and convex outer pole	3 (staggered circumferentially)	Mounted on rotary armature to produce rotation about stator axis	Custom NdFeB supplier
☐	BOM-04	High-Permeability Backing Plate (Linear)	Netic Co-Netic or equivalent μ -metal, ≥ 0.030 in thick, sized to cover all stator magnets	1	Concentrate stator magnetic flux; engages south poles of stator magnets	Magnetic Shield Corp. (Perfection Mica Co., Chicago)
☐	BOM-05	High-Permeability Annular Sleeve (Rotary)	Netic Co-Netic ring, concentric with stator axis, slides over stator cylindrical surface	1	Concentrate field at gaps between stator magnets in rotary embodiment	Magnetic Shield Corp.

✓	ID	PART	SPECIFICATION	QTY	PURPOSE	SOURCE
☐	BOM-13	Adhesive / Bonding Layer	Two-part epoxy (e.g., Loctite Epoxy Weld) or cyanoacrylate gel	1 tube	Accurately locate and fix stator magnets on backing plate; per patent disclosure	Hardware store
☐	BOM-14	Spacers (Stator Gap Shims)	Brass or aluminum shims in 0.060, 0.125, 0.188 in (varying per patent)	Set of 8+	Set slightly different spacing between adjacent stator magnets to smooth motion	McMaster-Carr shim stock
☐	BOM-15	Power Belt (Rotary)	Round neoprene belt, O-ring style, fits armature belt groove	1	Transmits rotary torque from armature to generator / load	McMaster-Carr
☐	BOM-16	Magnetization / Polarity Marker	Magnetic polarity indicator (e.g., Magnaflux pole finder)	1	Verify N/S orientation of every magnet before assembly	Magnaflux / K&J Magnetics
☐	BOM-17	Gaussmeter / Hall Probe	0–2000 G range, axial Hall probe (e.g., Lakeshore 410)	1	Quantify surface flux density; verify grain orientation	Lakeshore / AlphaLab
☐	BOM-18	Precision Calipers / Micrometer	0–6 in digital caliper (0.001 in resolution)	1	Set air gap to ~0.125 in and verify magnet dimensions	Mitutoyo / any
☐	BOM-19	Non-Magnetic Pin Vise / Puller	Brass or aluminum fixture for handling high-field magnets	1	Prevent pinch injuries; avoid ferrous tools near NdFeB magnets	Made / McMaster-Carr

Showing 19 of 19. Items marked **critical** cannot be substituted without re-validating the design.



Tools & Workshop Setup

What your workshop needs before you start cutting metal



Drill Press (Bench)

Precise hole drilling for fasteners and bearings

Required



Belt Sander / Disc Sander

Bevel armature pole ends; shape bowed profile

Required



Mini Mill or CNC Router

Cut slots in stator support plate; mill aluminum hub

Required



Diamond-coated Cutting Blade

Cut and shape NdFeB magnets (NEVER use steel blades)

Required



Tap & Die Set (M4 / #8)

Thread holes for stainless fasteners

Required



Vise (Non-Magnetic Jaws)

Hold aluminum and magnet work; brass jaw caps

Required



Digital Angle Gauge

Set bevel angles on armature ends ($\approx 15^\circ$)

Optional / Nice-to-have



Dial Indicator + Magnetic Base

Measure axial displacement for speed calibration

Optional / Nice-to-have



Stroboscope / Tachometer

Measure rotary RPM after build

Optional / Nice-to-have



Eye Protection + Gloves

NdFeB chips can shatter and fly when magnets snap together

Required



Linear Embodiment — Build Guide

7 phases from raw magnet stock to a moving prototype

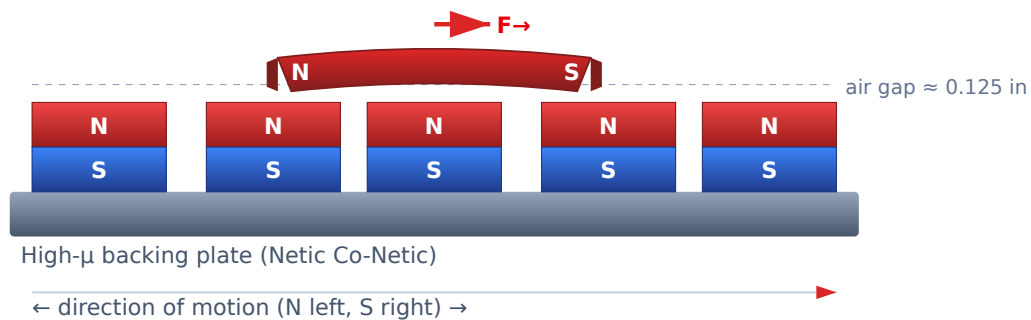
👁 Interactive Geometry

Drag the slider (or press Auto-run) to see how the armature traverses the stator track

Drag the slider (or press Auto-run) to see how the armature traverses the stator track

Each stator magnet (red = N up, blue = S down) is 1 in wide; gap is 0.25 in. Armature is 3 in long (≈ 2 stators + 1 gap).

Auto-run



Track start

Position: 50%

Track end

⚡ Always-On Thrust

At every position, the N pole of the armature is repelled by the next stator N pole while the S pole is attracted by the previous stator S pole. Net force vector always points left.

📏 2-Stator Length Ratio

Armature length ≈ 2 stator widths + 1 gap. This guarantees that one end is always in repulsion while the other is always in attraction.

⚙ Variable Stator Spacing

Stator gaps alternate slightly to smooth the force pulse. Reversing the armature (N $\leftrightarrow$ S) reverses the direction of motion.

Build checklist progress

0 / 28 checks

1

Preparation

Inventory & Polity Verification

0/4 checks ▾

Est. duration: 1–2 h

Lay out every NdFeB magnet on a non-magnetic workbench. Use a polarity indicator to label N and S on every stator and armature magnet with a paint marker. Mismatched polarity is the #1 cause of a non-functional build, so verify every single magnet before bonding anything.

**Warning**

NdFeB magnets can pinch and shatter. Wear eye protection and use a non-magnetic vise. Never let two large magnets snap together uncontrolled.

VERIFICATION CHECKLIST

- ☐ All stator magnets have N facing the same direction (toward future air gap)
- ☐ Armature magnet poles are clearly marked at the ENDS (not the broad faces)
- ☐ Backup plate (Netic Co-Netic) is verified non-magnetic in its as-supplied state
- ☐ Gaussmeter readings on every stator magnet are within $\pm 5\%$ of nominal

2

Foundation

Mount Backing Plate to Stator Support

0/3 checks ▾

Est. duration: 1 h

Bond the Netic Co-Netic backing plate to the stator support plate using thin film epoxy. Clamp evenly across the entire surface; an uneven bond creates air pockets that act as flux barriers. Cure for the full epoxy setting time before proceeding.

**Warning**

Do not heat-cure above 80 °C — μ -metal can lose its permeability if heated excessively.

VERIFICATION CHECKLIST

- ☐ Backing plate is flush with the support plate; no gaps
- ☐ Backing plate covers the entire area where stator magnets will sit
- ☐ Cured at room temperature for the full manufacturer-recommended time

Position the first stator magnet with its SOUTH pole facing the backing plate (north pole facing up, where the armature will run). Apply a thin, even layer of epoxy to the south face and press firmly onto the backing plate. Use a spacing jig to set the gap to the next magnet.

**Warning**

If any magnet snaps onto the backing plate uncontrolled, it can chip or take skin with it. Use brass spacers as a temporary mechanical stop.

VERIFICATION CHECKLIST

- ☐ All stator magnets have their NORTH poles facing up (toward the air gap)
- ☐ Gaps alternate slightly (e.g., 0.250, 0.3125, 0.250, 0.3125 in) per the patent's variable-spacing principle
- ☐ Magnets are aligned flush at the front edge; no magnet is canted in any axis
- ☐ Epoxy is fully cured before removing the alignment jig

4

Armature

Shape the Armature Magnet

0/4 checks ▾

Est. duration: 3–5 h

Cut the armature bar to 3.000 in length using a diamond-coated blade (NEVER use steel — it will chip the magnet and contaminate the blade). Sand the broad faces to a gentle arc (bow ~ 0.030 – 0.060 in over the length) using a belt sander with a contoured platen. Bevel the pole ends to $\sim 15^\circ$ to reduce cross-section at the tip.



Warning

Dry-cutting NdFeB produces pyrophoric dust that can ignite. Use water-cooled cutting and a dust collection system. Wear a particulate respirator.

VERIFICATION CHECKLIST

- ☐ Total length is 3.000 in ± 0.005 in
- ☐ Bowed profile is concave-down toward the stator track
- ☐ Bevel angle on both ends is symmetric ($\sim 15^\circ$)
- ☐ Surface is smooth; no chips or cracks (chips create flux leakage points)

5

Armature

Mount Armature to Carriage

0/4 checks ▾

Est. duration: 2 h

Bond the armature magnet (concave side down) to the non-magnetic carriage using epoxy. The carriage slides on linear rails that are mounted to the same base plate as the stator — the rail axis is exactly parallel to the stator track direction.

VERIFICATION CHECKLIST

- ☐ Carriage moves freely along the entire 6+ in travel with no binding
- ☐ Air gap (carriage bottom $\leftrightarrow$ stator north face) is 0.125 in ± 0.005 in
- ☐ Armature pole axis is perpendicular to the track direction
- ☐ Carriage cannot lift off the rails (vertical constraint present)

6

Tuning

Initial Test — Static Pull-Force Measurement

0/3 checks ▾

Est. duration: 1 h

Lock the carriage at one end of the track. Attach a spring scale to the carriage and pull slowly along the track direction. Record force at multiple positions; you should observe a non-zero bias force in one direction. If the force oscillates symmetrically about zero, the polarity of the armature or the stator magnets is wrong — re-verify with the polarity indicator.

**Warning**

If force reverses direction at any point in the travel, your armature is mounted upside down (poles swapped). Disassemble and re-bond.

VERIFICATION CHECKLIST

- ☐ Net force is biased in the intended direction of motion at every position
- ☐ Force magnitude is at least 0.5 lbf for a single-magnet prototype
- ☐ No audible clicking from magnets shifting in their bonds

7

Tuning

Free-Run & Smoothing

0/3 checks ▾

Est. duration: 2 h

Release the carriage and observe motion. If the carriage stops partway, increase the air gap by 0.010 in and retry; if motion is jerky, install a second staggered armature magnet and/or vary the stator gaps further. Iterate until the carriage traverses the full track smoothly.

VERIFICATION CHECKLIST

- ☐ Carriage completes the full track length on a single release
- ☐ Motion is smooth with no perceptible pulsation
- ☐ Adding a load (small weight on a string) shows measurable thrust



Rotary Embodiment — Build Guide

7 phases for the circular motor with speed regulator

👁 Interactive Geometry

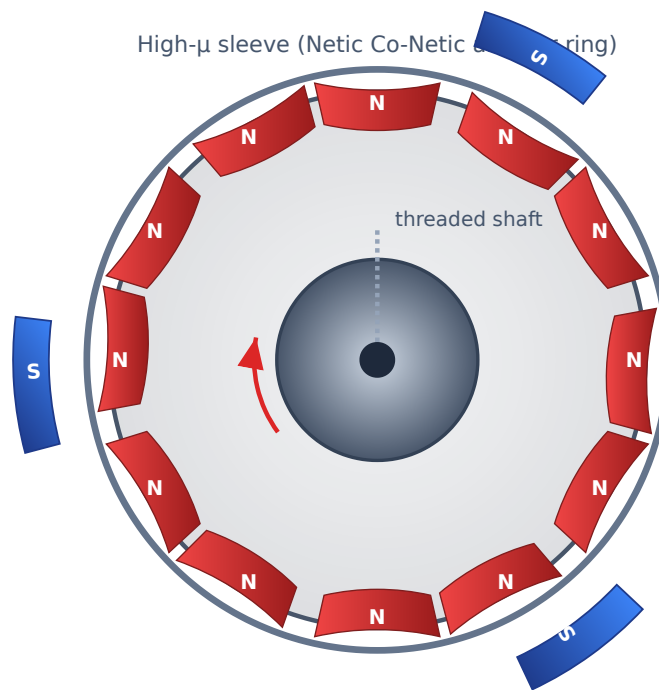
Watch the 3 staggered armature magnets rotate around 12 stator magnets — adjust axial engagement to change speed

Interactive rotary motor — 3 staggered armature magnets rotate around 12 stator magnets

Watch the 3 staggered armature magnets rotate around 12 stator magnets — adjust axial engagement to change speed

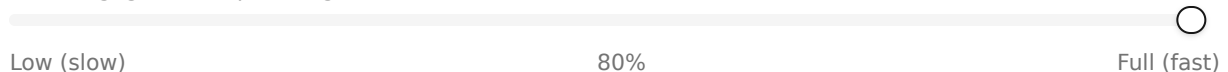
Pause

Axial engagement: 80%



↻ rotation (staggered armature magnets 0°, 122°, 244°)

Axial engagement (speed regulator)



Build checklist progress

0 / 28 checks

1

Preparation

Verify Wedge Magnet Set

0/3 checks ▾

Est. duration: 1 h

The 3 rotary armature magnets are wedge-shaped with a curved inner surface and convex outer pole. Verify that all three have identical wedge angles and that their inner curves match the outer radius of the stator sleeve.

VERIFICATION CHECKLIST

- ☐ All 3 wedge magnets have the same wedge angle
- ☐ Inner curved surfaces have the same radius
- ☐ Polarity is verified with N facing radially outward (or inward, consistently)

2

Foundation

Machine Stator Cylinder

0/3 checks ▾

Est. duration: 3-4 h

Turn the non-magnetic stator cylinder (aluminum or Delrin) to 6 in OD × 2 in height. Bore the concentric threaded hole (1/4-20) through the center. Cut the annular groove on the outer cylindrical surface to receive the high-permeability sleeve.

VERIFICATION CHECKLIST

- ☐ Outer cylinder is concentric with the threaded bore within 0.002 in TIR
- ☐ Annular groove depth matches sleeve thickness
- ☐ Threaded bore is clean and runs smoothly with the shaft

3

Foundation

Install High- μ Sleeve & Stator Magnets

0/4 checks ▾

Est. duration: 2–3 h

Bond the Netic Co-Netic sleeve into the annular groove. Then bond each stator magnet (wedge profile, curved inner surface) onto the sleeve with its south pole facing the sleeve. Space them around the circumference with slight variations to smooth rotation.

**Warning**

A radial misalignment of one stator magnet will produce a strong angular pulsation. Use a fixture to hold each magnet in place during cure.

VERIFICATION CHECKLIST

- ☐ Sleeve is fully seated with no gaps
- ☐ All stator wedges have N facing radially outward
- ☐ Spacing varies by ~10% between adjacent magnets
- ☐ Outer pole surfaces form a smooth circular locus

4

Armature

Build the Dished Armature

0/3 checks ▾

Est. duration: 3 h

Machine the dished armature from aluminum or Delrin: a radial web with an axially extending portion that holds the 3 wedge magnets. Cut a belt-receiving groove on the outer rim. Drill and tap holes for magnet fastening.

VERIFICATION CHECKLIST

- ☐ Armature is concentric with the stator within 0.005 in TIR
- ☐ Belt groove is properly sized for the chosen belt
- ☐ Three magnet mounting pads are at the correct stagger angles (NOT 120° — slight asymmetry is intentional)

5

Armature

Mount Wedge Magnets with Stagger

0/3 checks ▾

Est. duration: 2 h

Bond each wedge magnet to its mounting pad with the curved inner face toward the stator. Stagger them circumferentially so they are NOT at 120° — for example at 0°, 122°, 244°. This stagger smooths the tangential force as the armature rotates.



Warning

If two wedge magnets are misaligned by even 5°, the rotational force will have a noticeable angular pulsation that wastes energy.

VERIFICATION CHECKLIST

- ☐ All 3 magnets are at the correct stagger angles (verified with protractor)
- ☐ All have the same polarity facing the stator
- ☐ Air gap between each magnet and the stator poles is uniform (~0.125 in)

6

Assembly

Install Threaded Shaft & Bearings

0/3 checks ▾

Est. duration: 1.5 h

Press two antifriction ball bearings into the armature hub. Slide the threaded shaft through the bearings and thread it into the stator bore. Attach the adjustment knob to the outer end of the shaft. The armature should now spin freely and also move axially when the knob is rotated.

VERIFICATION CHECKLIST

- ☐ Armature rotates freely with no radial play
- ☐ Turning the knob moves the armature axially along the shaft
- ☐ Axial travel range is at least 0.5 in for usable speed control

Install the belt around the armature groove and connect it to a small generator or flywheel as a load. Release the armature with full axial engagement. Measure RPM with a stroboscope. Adjust the knob to find the maximum-RPM engagement and the minimum-RPM engagement (just before stall).

**Warning**

If the armature oscillates back-and-forth instead of rotating continuously, your stator magnet spacing is too uniform. Vary it by 10-15% to break the symmetry.

VERIFICATION CHECKLIST

- ☐ Armature rotates continuously in one direction when released
- ☐ Speed changes smoothly with knob rotation
- ☐ Measured maximum RPM is at least 100 with no load (typical for a small prototype)
- ☐ Belt tension is sufficient to prevent slip under load

🔧 Force vs Air-Gap Simulator

A first-order engineering model to help you choose the right air gap and magnet grade for your build

Air gap: 0.125 in (3.17 mm)

0.050 in

0.300 in

Magnet grade: N50

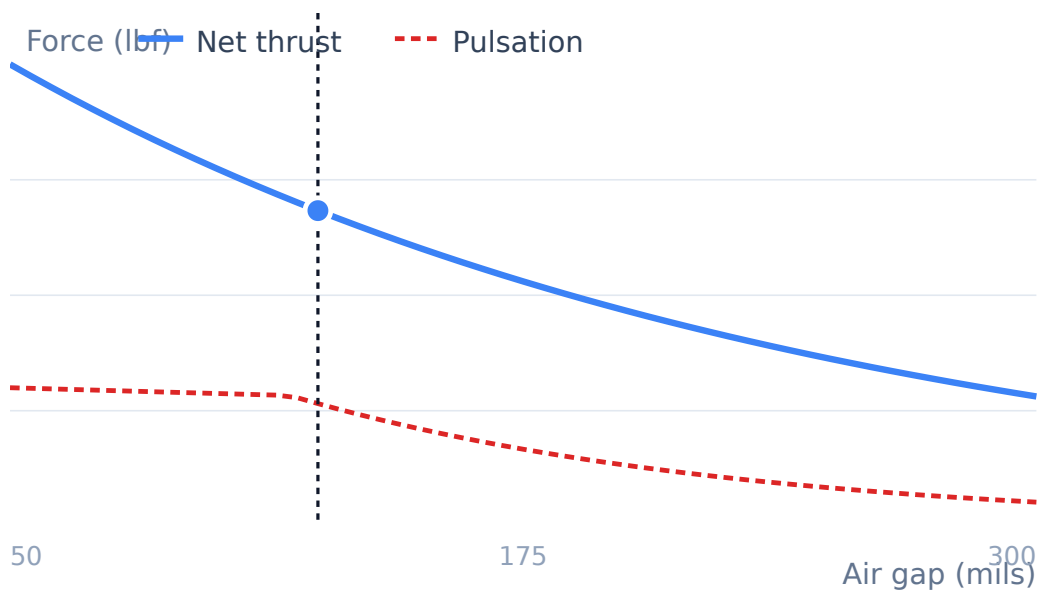
N35

N55

🔧 Estimated Force Output

Net thrust (lbf)	4.69
Peak pulsation (lbf)	1.82
Smoothness index	61%
Magnet remanence Br	1.33 T

📊 Trade-off Visualization



Blue = net thrust vs air gap; red = pulsation amplitude. The patent recommends an air gap around 0.125 in (3.18 mm) — close to the "knee" where thrust is still high but pulsation begins to drop off.



Optimization Tip

Acceptable; consider widening gap slightly to reduce pulsation.

Model: $F = k \cdot Br^2 \cdot \exp(-\text{gap} / 5\text{mm})$. Pulsation approximated as a Gaussian peak around the optimal gap. Values are illustrative — your actual build will differ based on magnet quality, geometry tolerances, and surface finish.



Safety, Warnings & Engineering Notes

Critical information before you handle high-field magnets

Pinch & Shatter Hazard critical

NdFeB magnets above N40 grade generate forces exceeding 50 lbf when in direct contact. A sudden snap can shatter the magnet, sending sharp chips flying at high velocity. Always wear eye protection and use non-ferrous tooling (brass, aluminum, or polymer) when handling.

Pyrophoric Machining Dust critical

Dry-machining NdFeB produces a fine iron-neodymium dust that is pyrophoric — it can ignite spontaneously in air. Always use water-cooled diamond cutting and a dust extraction system. Do not accumulate dust in a bin; flush with water immediately.

Magnetic Field Interference caution

Keep the assembled motor at least 12 in (30 cm) away from pacemakers, magnetic storage media, CRT displays, and credit-card stripes. Static fields from N52 magnets can wipe magnetic stripe cards at distances up to 6 in.

Heat Sensitivity of NdFeB caution

NdFeB magnets begin to lose magnetization irreversibly above 80 °C. Do not heat-cure adhesives near the magnets; use room-temperature epoxy. Store magnets below 60 °C.

High-Permeability Alloy Handling note

Netic Co-Netic and other μ -metal alloys are mechanically soft and dent easily. Do not bend or hammer them — even a small dent creates a local stress that reduces permeability and degrades the field-concentration effect.

Scientific Status of the Invention note

The patent describes a 'motive power source solely through superconducting characteristics of a permanent magnet.' This is not accepted by mainstream physics; the second law of thermodynamics implies a permanent magnet alone cannot produce net work indefinitely. The patent is best treated as an engineering case study of magnetic-field manipulation, not a blueprint for a working free-energy machine.



Patent Claims — Reference

The 25 legal claims that define the invention's scope



1

apparatus

independent

A permanent magnet motor comprising a stator track defining a track direction and having first and second sides composed of a plurality of track permanent magnets each having first and second poles of opposite polarity, said magnets being disposed in side-by-side relationship with a spacing between adjacent magnets and like poles defining said track sides, an elongated armature permanent magnet located on one of said track sides for relative movement thereto in spaced relationship, said armature magnet having first and second poles of opposite polarity located at the opposite ends defining the length thereof, the length of said armature magnet being disposed in general alignment with the direction of said track, the spacing of said armature magnet poles from said track side and the length of said armature magnet as related to the width and spacing of said track magnets in the direction of said track being such as to impose a continuous force on said armature magnet in said general direction of said track.

2

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein the spacing between said poles of said armature magnet and the adjacent stator track side are substantially equal.

3

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein the spacing between adjacent track magnets varies.

4

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein a plurality of armature magnets are disposed on a common side of said stator track, said armature magnets being mechanically interconnected.

5

subcombination

dependent

In a permanent magnet motor as in claim 4 wherein said armature magnets are staggered with respect to each other in the direction of said track.

6

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein magnetic field concentrating means are associated with said track magnets.

7

subcombination

dependent

In a permanent magnet motor as in claim 6 wherein said field concentrating means comprises a sheet of magnetic material of high magnetic field permeability engaging side and pole of said track magnets opposite to that side and pole disposed toward said armature magnet.

8

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein said armature magnet is of an arcuate configuration in its longitudinal direction bowed toward said track, said armature magnet having ends shaped to concentrate the magnetic field at said ends.

9

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein said stator track is of a generally linear configuration, and means supporting said armature magnet relative to said track for generally linear movement of said armature magnet.

10

subcombination

dependent

In a permanent magnet motor as in claim 1 wherein said stator track magnets define a circle having an axis, an armature rotatably mounted with respect to said track and concentric and coaxial thereto, said armature magnet being mounted upon said armature.

11

subcombination

dependent

In a permanent magnet motor as in claim 10, means axially adjusting said armature relative to said track whereby the axial relationship of said armature magnet and said stator magnets may be varied to adjust the rate of rotation of said armature.

12

subcombination

dependent

In a permanent magnet motor as in claim 10 wherein a plurality of armature magnets are mounted on said armature.

13

subcombination

dependent

In a permanent magnet motor as in claim 12 wherein said armature magnets are circumferentially nonuniformly spaced on said armature.

14

apparatus

independent

A permanent magnet motor comprising a stator comprising a plurality of circumferentially spaced stator permanent magnets having poles of opposite polarity, said magnets being arranged to substantially define a circle having an axis, the poles of said magnets facing in a radial direction with respect to said axis and poles of the same polarity facing away from said axis and the poles of opposite polarity facing toward said axis, an armature mounted for rotation about said axis and disposed adjacent said stator, at least one armature permanent magnet having poles of opposite polarity mounted on said armature and in radial spaced relationship to said circle of stator magnets, said armature magnet poles extending in the circumferential direction of armature rotation, the spacing of said armature magnet poles from said stator magnets and the circumferential length of said armature magnet and the spacing of said stator magnets being such as to impose a continuing circumferential force on said armature magnet to rotate said armature.

15

subcombination

dependent

In a permanent magnet motor as in claim 14 wherein a plurality of armature magnets are mounted upon said armature.

16

subcombination

dependent

In a permanent magnet motor as in claim 14 wherein said armature magnets are asymmetrically circumferentially spaced on said armature.

17

subcombination

dependent

In a permanent magnet motor as in claim 14 wherein the poles of said armature magnet are shaped to concentrate the magnetic field thereof.

18

subcombination

dependent

In a permanent magnet motor as in claim 14, magnetic field concentrating means associated with said stator magnets concentrating the magnetic fields thereof at the spacings between adjacent stator magnets.

19

subcombination

dependent

In a permanent magnet motor as in claim 18 wherein said magnet field concentrating means comprises an annular ring of high magnetic field permeability material concentric with said axis and in substantial engagement with poles of like polarity of said stator magnets.

20

subcombination

dependent

In a permanent magnet motor as in claim 14 wherein said armature magnet is of an arcuate bowed configuration in the direction of said poles thereof defining a concave side and a convex side, said concave side being disposed toward said axis, and said poles of said armature magnet being shaped to concentrate the magnetic field between said poles thereof.

21

subcombination

dependent

In a permanent magnet motor as in claim 14, means for axially displacing said stator and armature relative to each other to adjust the axial alignment of said stator and armature magnets.

22

method

independent

The method of producing a unidirectional motive force by permanent magnets using a plurality of spaced stator permanent magnets having opposite polarity poles defining a track having a predetermined direction, and an armature magnet having a length defined by poles of opposite polarity movably mounted for movement relative to the track in the direction thereof, and of a predetermined length determined by the width and dimensions of said stator magnets comprising forming a magnetic field track by said stator magnets having a magnetic field of common polarity interrupted at spaced locations in a direction transverse to the direction of said magnetic field track by magnetic fields created by magnetic lines of force existing between the poles of the stator magnets and positioning the armature magnet in spaced relation to said magnetic field track longitudinally related to the direction of the magnetic field track such a distance that the repulsion and attraction forces imposed on the armature magnet by said magnetic field track imposes a continuing unidirectional force on the armature magnet in the direction of the magnetic field track.

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method

dependent

The method of producing a unidirectional motive force as in claim 22 including concentrating the magnetic fields created by magnetic lines of force between the poles of the stator magnets.

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method

dependent

The method of producing a unidirectional motive force as in claim 22 including concentrating the magnetic field existing between the poles of the armature magnet.

25

method

dependent

The method of producing a unidirectional motive force as in claim 22 including concentrating the magnetic fields created by magnetic lines of force between the poles of the stator magnets and concentrating the magnetic field existing between the poles of the armature magnet.



Troubleshooting Guide

Common failure modes and how to fix them

⚠️ Armature does not move at all when released

LIKELY CAUSE

Stator magnets have mixed polarity, or armature is mounted upside down

FIX

Re-verify polarity of every stator magnet with a polarity indicator. Confirm armature N/S orientation matches intended direction.

⚠️ Armature moves but stalls partway down the track

LIKELY CAUSE

Air gap is too small at one position (local pulsation overwhelms the net force), or stator magnets are too uniform

FIX

Increase air gap by 0.010 in increments. Vary stator gaps by $\pm 10\%$ (e.g., 0.225, 0.275, 0.225, 0.275 in).

⚠️ Motion is jerky / pulsating

LIKELY CAUSE

Single armature magnet sees each stator gap as a discrete event

FIX

Add a second armature magnet, staggered by $\sim 50\%$ of one stator pitch. Or vary stator spacing more aggressively.

⚠️ Rotary armature oscillates back-and-forth instead of rotating

LIKELY CAUSE

Stator magnet spacing is perfectly uniform — the armature finds a symmetric minimum-energy oscillation

FIX

Re-bond one stator magnet with a 10% different gap. Break the symmetry deliberately.

 Speed drops sharply when a small load is applied

LIKELY CAUSE

Air gap is too large, or axial engagement of rotary armature is too low

FIX

Reduce air gap to ~0.100 in. Rotate the knob to increase axial engagement to ~80% of full.

 Magnets shift on the backing plate after running

LIKELY CAUSE

Epoxy bond was insufficient or magnets were assembled before full cure

FIX

Disassemble, clean all surfaces with acetone, re-bond with high-strength epoxy, and cure for the full recommended time before handling.